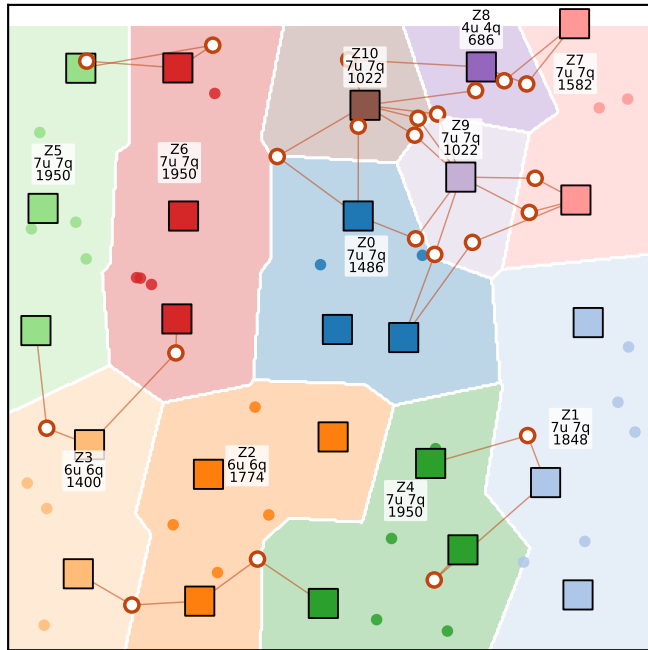
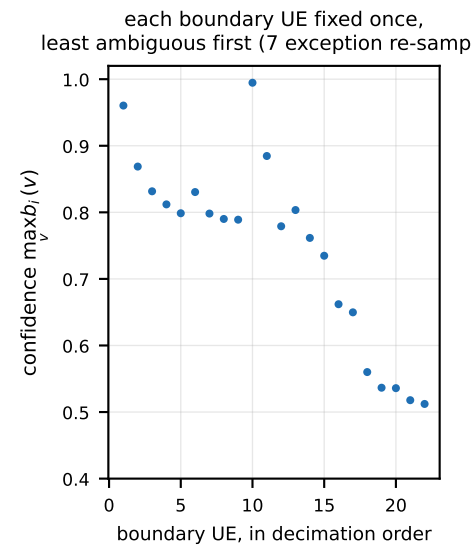
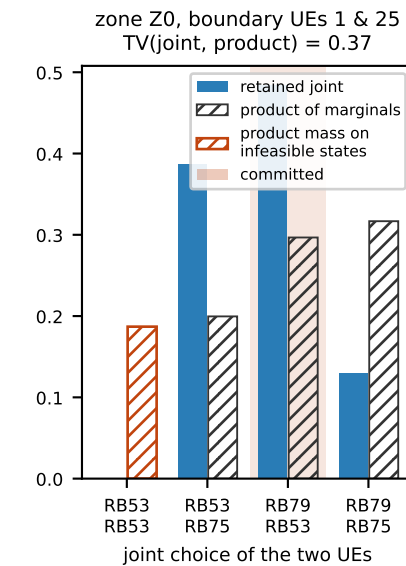
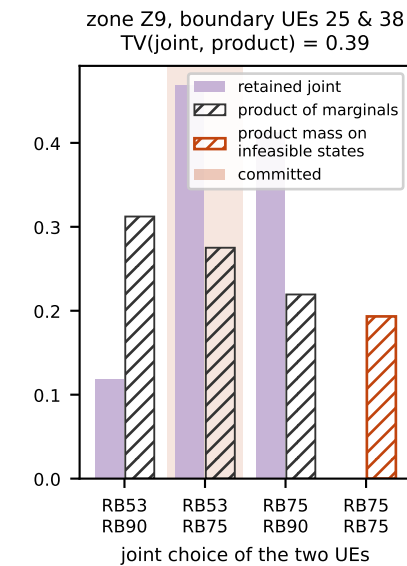
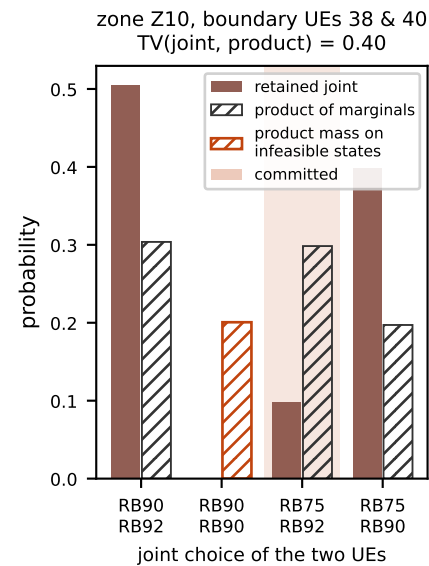
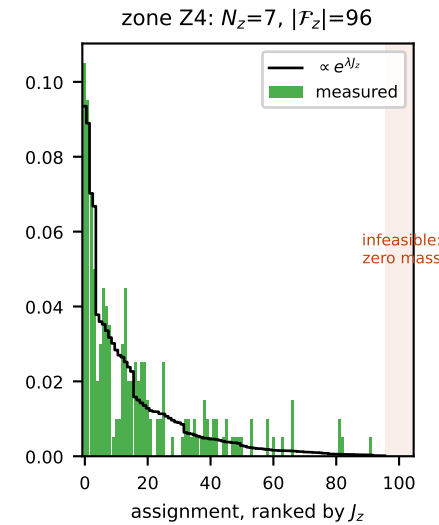
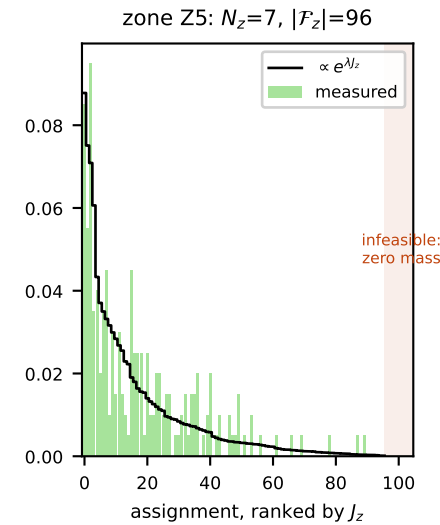
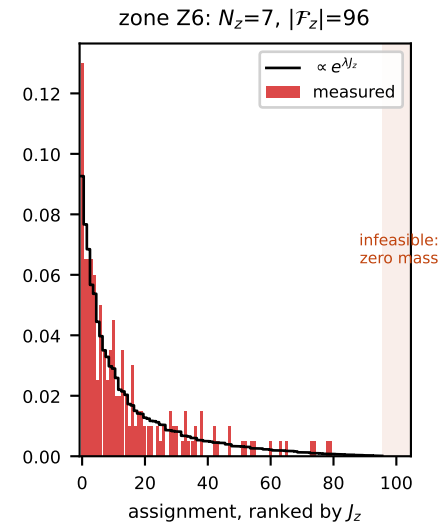
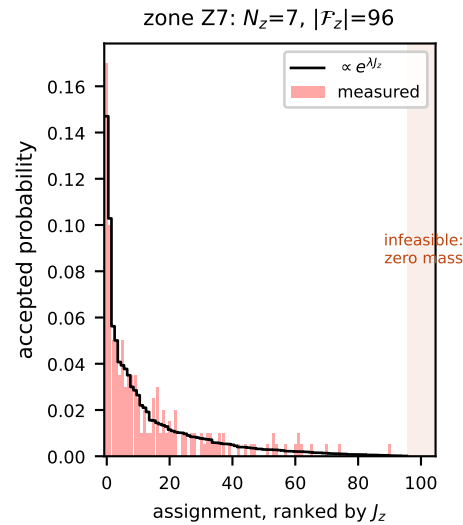


A region partitioned to the hardware budget, solved by zone-local sampling and classical boundary coordination

(a) partition under a 2000 two-qubit budget
25 APs, 50 UEs $\rightarrow$ 11 zones, $|B|=22$ boundary UEs (44%)



shading: zone territory (nearest AP)
 ■ AP (colour = zone) ● interior UE
 ○ boundary UE — UE-to-candidate-AP link



$g=5$, seed=7, chosen for legibility; its utility ratio sits at the mean of 12 candidate seeds (99.3%, spread 97.7-100.0%). $\beta=1.5$, $K_z=200$. Utility 276.4 of the centralized strict optimum 278.7 = 99.2%; the assignment is strictly feasible (True). Zone labels in (a): UEs, state quo
 Zone laws are exact: the accepted branch of the circuit matches $\exp(\lambda J_z)$ on F_z to statevector precision (max TVD $2.7e-15$ over the zones checked in `haiq_certify.py`), so the sampler used here is the same law the circuit produces.